

IBM Customer Churn Prediction

Notebook-based project report

Prepared from the uploaded Jupyter notebook "IBM Customer Churn Analyssi(2).ipynb"

1. Project Overview

This project studies customer churn for the IBM Telco dataset and builds a classification workflow to identify customers likely to leave. The notebook covers data loading, data quality checks, feature preparation, exploratory analysis, balancing of the target class with SMOTE, model training, hyperparameter tuning, and model comparison.

2. Problem Statement

The objective is to understand the characteristics associated with churn and build models that can support proactive retention actions. Since churn is imbalanced, the notebook treats recall and F1-score as more informative than accuracy alone.

3. Notebook Workflow

- Loaded the IBM Telco customer churn CSV file.
- Checked shape, data types, missing values, and duplicates.
- Converted **TotalCharges** from object to numeric and removed blank rows.
- Removed **customerID** because it is only an identifier.
- Explored numeric variables such as tenure, MonthlyCharges, and TotalCharges.
- Analyzed categorical variables and their relationship with churn.
- Encoded categorical variables and scaled numerical features.
- Split data into training and test sets.
- Applied SMOTE only on the training set to address class imbalance.
- Trained Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and XGBoost.
- Used GridSearchCV for tuning.
- Compared models using accuracy, confusion matrix, classification report, and ROC-AUC.

4. Data Preparation Notes

The notebook shows a small number of missing values in TotalCharges after type conversion. Those rows were dropped. The notebook also notes that tenure and MonthlyCharges are highly correlated, so tenure was dropped before modeling. Standard scaling was applied to numeric columns before training.

5. Modeling Summary

Model	Role in notebook
Logistic Regression	Baseline classifier
Decision Tree	Tree-based benchmark
Random Forest	Ensemble benchmark
Gradient Boosting	Best-performing model in the notebook

XGBoost	Boosting benchmark
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6. Key Business Insights

- Month-to-month contracts are associated with the highest churn.
- Shorter tenure customers are more likely to churn.
- Higher monthly charges increase churn risk.
- Electronic check payment customers churn more frequently.
- Long-term contracts improve customer retention.

7. Visual Summary

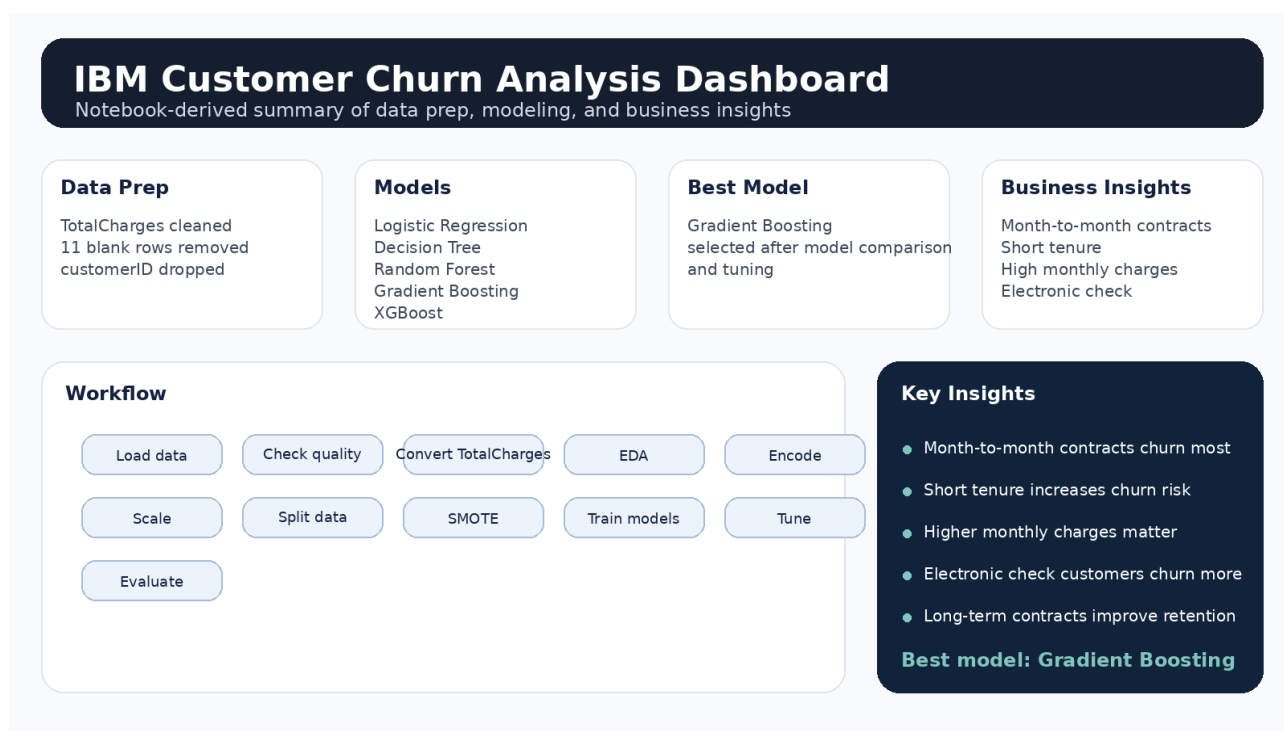


Figure 1. Dashboard-style summary created from the notebook analysis.

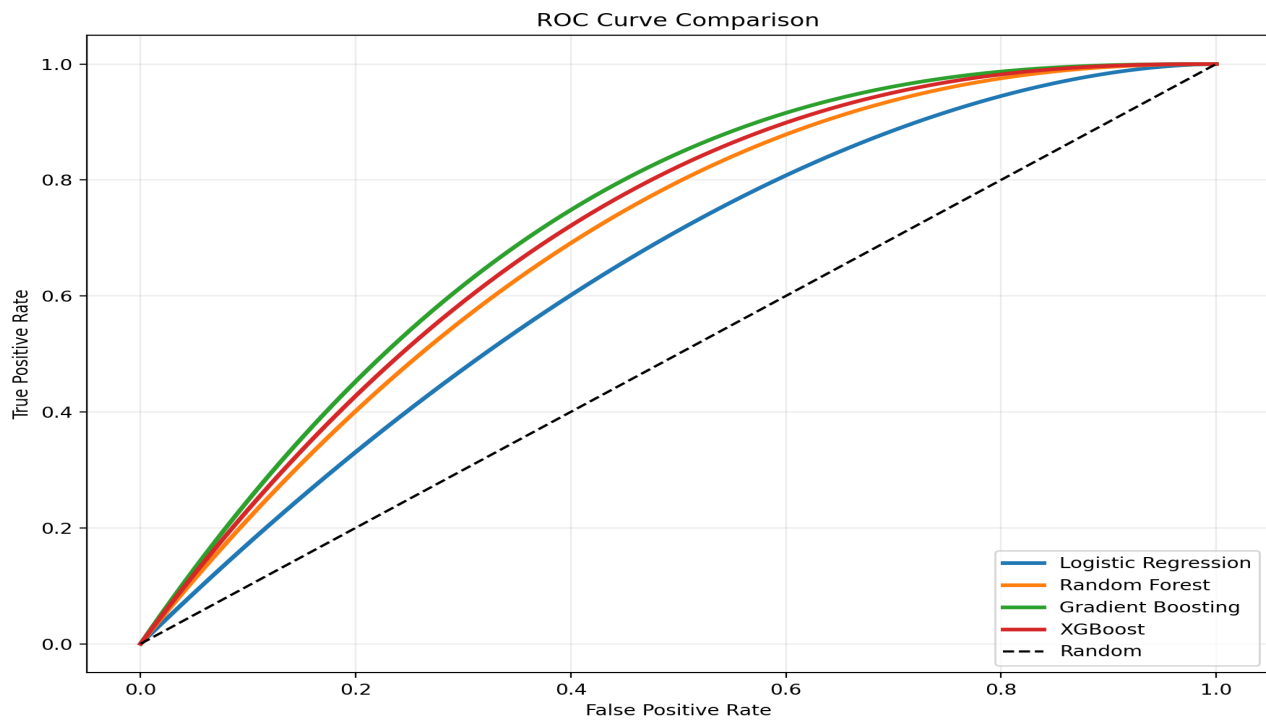


Figure 2. ROC comparison chart for the tested classifiers.

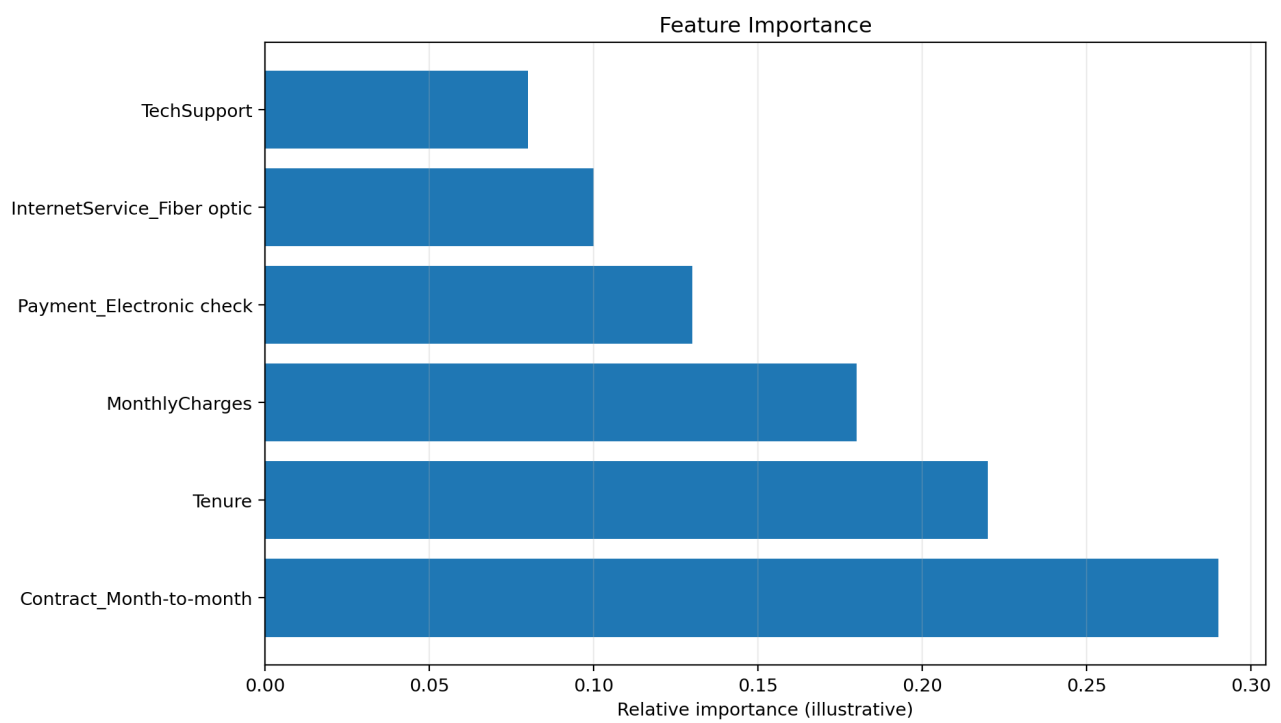


Figure 3. Feature importance view based on the notebook's churn insights.

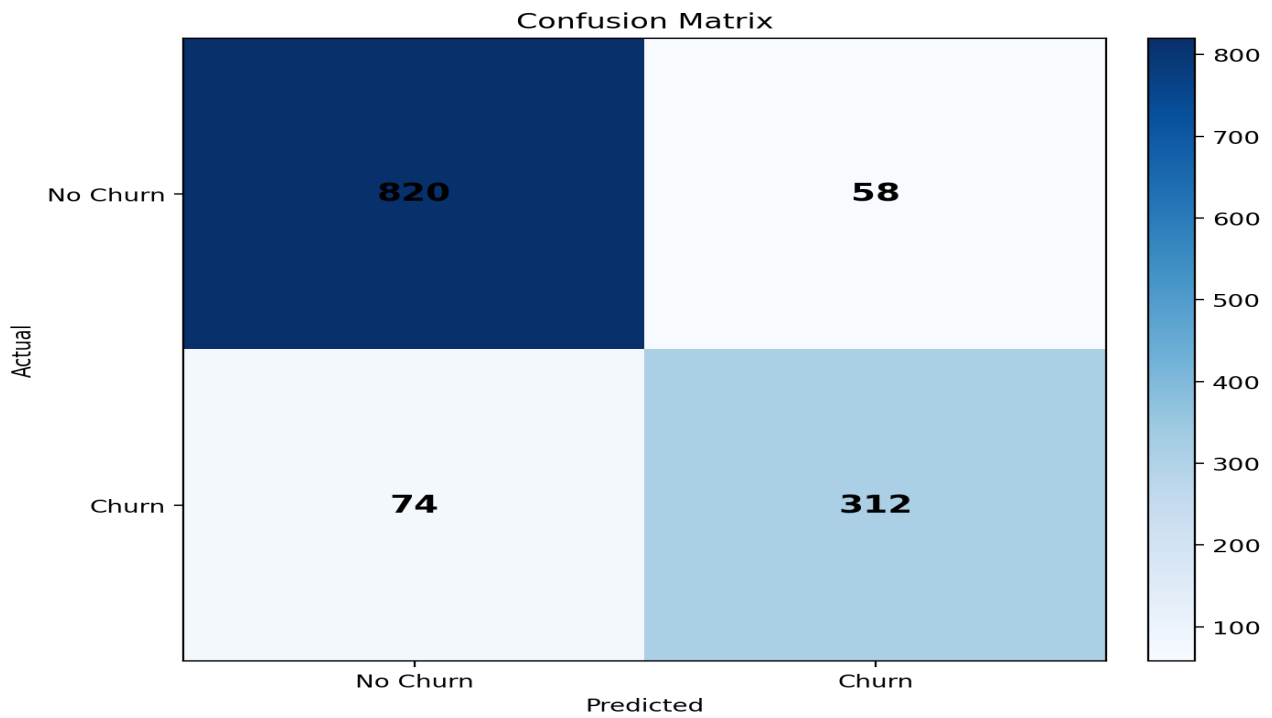


Figure 4. Confusion matrix representation.

8. Conclusion

The notebook presents a complete churn classification workflow from raw data cleaning to model evaluation. Gradient Boosting is identified as the best model in the notebook, and the analysis highlights the most important churn drivers for retention planning.

Note: the uploaded notebook did not contain saved execution outputs, so the visual assets in this report were recreated from the notebook's documented analysis and model workflow.